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CS-F

Homework 3:

Problem 1:

(a) $x_n = x_{n-1} + x_{n-2}$

$a_0 = 1$

$a_1 = 1$

$$C = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$$

(b) $x_n = 4x_{n-1} - x_{n-2} - 6x_{n-3}$

$a_0 = -6$

$a_1 = -1$

$a_2 = 4$

$$C = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -1 & 4 \end{bmatrix}$$

(c) $x_n = x_{n-1} + 4x_{n-3} + 2x_{n-4}$

$a_0 = 2$

$a_1 = 4$

$a_2 = 0$

$a_3 = 1$

$$C = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 2 & 4 & 0 & 1 \end{bmatrix}$$

Problem 2:

$x_0 = 3, \; x_1 = -2, \; x_2 = 8$

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -1 & 4 \end{bmatrix}^n \begin{bmatrix} 3 \\ -2 \\ 8 \end{bmatrix}$$

Calculating eigenvalues:

$|A - \lambda I| = 0$

$$= \left| \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -1 & 4 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} \right| = 0$$

$$= \left| \begin{bmatrix} -\lambda & 1 & 0 \\ 0 & -\lambda & 1 \\ -6 & -1 & 4-\lambda \end{bmatrix} \right|$$

$$-(\lambda - 3)(\lambda - 2)(\lambda + 1) = 0$$

$\lambda_3 = 3, \; \lambda_2 = 2, \; \lambda_1 = -1$

For eigenvectors:

For $\lambda = 3$

$$\begin{bmatrix} \cancel{-\lambda}-3 & 1 & 0 \\ 0 & -3 & 1 \\ -6 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

After row reduction

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & -1/9 & 0 \\ 0 & 1 & -1/3 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\text{Span } \left\{ \begin{bmatrix} 1/9 \\ 1/3 \\ 1 \end{bmatrix} \right\}$$

$$\text{Span } \left\{ \begin{bmatrix} 9 \\ 3 \\ 9 \end{bmatrix} \right\}$$

For $\lambda = 2$

$$\begin{bmatrix} -2 & 1 & 6 \\ 0 & -2 & 1 \\ -6 & -1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

After row reduction

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & -1/4 & 0 \\ 0 & 1 & -1/2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\text{Span} = \left\{ \begin{bmatrix} 1/4 \\ 1/2 \\ 1 \end{bmatrix} \right\}$$

$$\text{Span} = \left\{ \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} \right\}$$

For $\lambda = -1$

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ -6 & -1 & 5 \end{bmatrix}.$$

After row reduction

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\text{span} = \left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}$$

$$D = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{bmatrix}, \quad P = \begin{bmatrix} 1 & 1 & 1 \\ 3 & 2 & -1 \\ 9 & 4 & 1 \end{bmatrix}$$

$$P^{-1} = \frac{1}{12} \begin{bmatrix} -6 & -3 & 3 \\ 12 & 8 & -4 \\ 6 & -5 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -1 & 4 \end{bmatrix}^n \begin{bmatrix} 3 \\ -2 \\ 8 \end{bmatrix} = PD^nP^{-1} \begin{bmatrix} 3 \\ -2 \\ 8 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 \\ 3 & 2 & -1 \\ 9 & 4 & 1 \end{bmatrix} \begin{bmatrix} 3^n & 0 & 0 \\ 0 & 2^n & 0 \\ 0 & 0 & (-1)^n \end{bmatrix} \left(\frac{1}{12} \begin{bmatrix} -6 & -3 & 3 \\ 12 & 8 & -4 \\ 6 & -5 & 1 \end{bmatrix} \right) \begin{bmatrix} 3 \\ -2 \\ 8 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 \\ 3 & 2 & -1 \\ 9 & 4 & 1 \end{bmatrix} \begin{bmatrix} 3^n & 0 & 0 \\ 0 & 2^n & 0 \\ 0 & 0 & (-1)^n \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 \\ 3 & 2 & -1 \\ 9 & 4 & 1 \end{bmatrix} \begin{bmatrix} 3^n \\ -2^n \\ 3(-1)^n \end{bmatrix}$$

Thus, $3^n - 2^n + 3(-1)^n$

$$\boxed{x_n = 3^n - 2^n + 3(-1)^n}$$